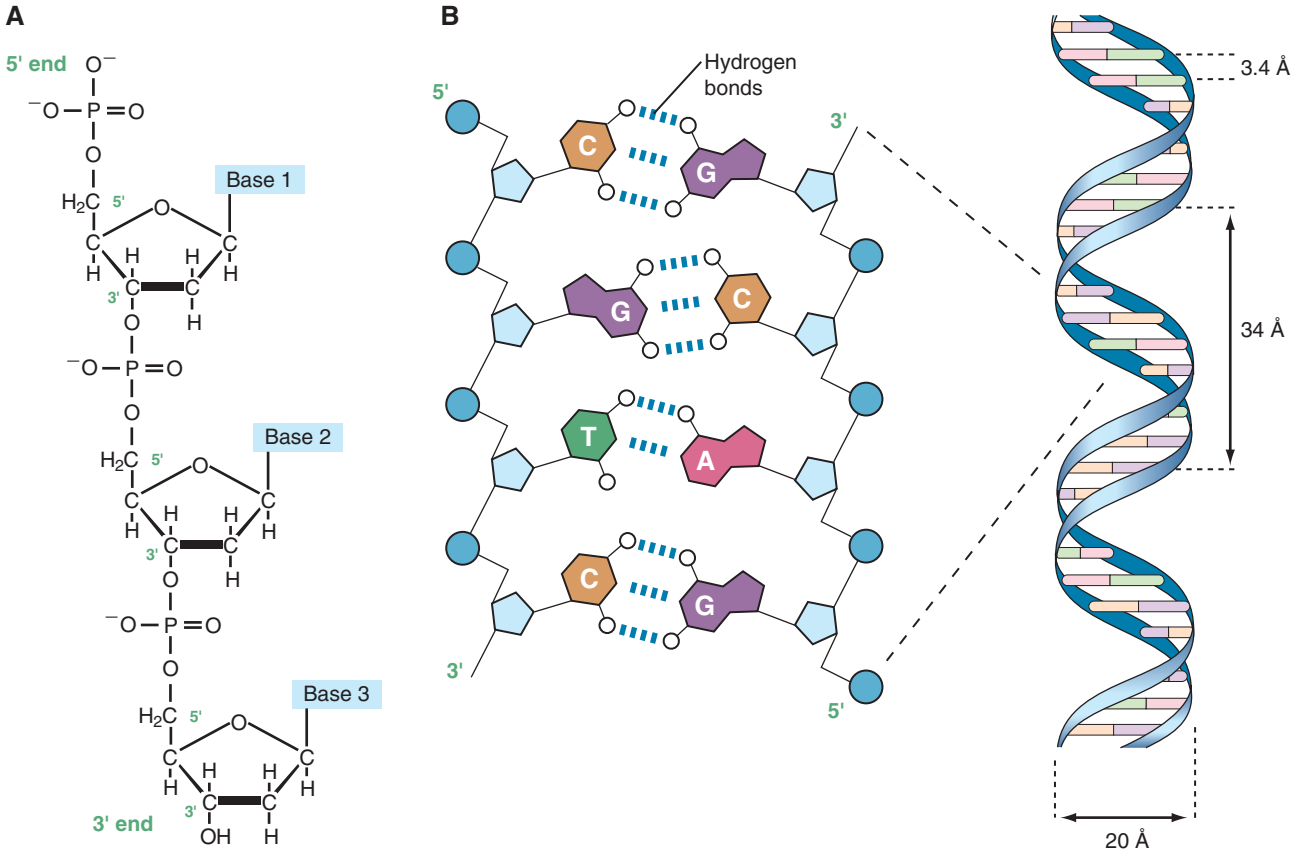


Figure 2-3 **The structure of DNA. A,** A portion of a DNA polynucleotide chain, showing the 3′-5′ phosphodiester bonds that link adjacent nucleotides. **B,** The double-helix model of DNA, as proposed by Watson and Crick. The horizontal "rungs" represent the paired bases. The helix is said to be right-handed because the strand going from lower left to upper right crosses over the opposite strand. The detailed portion of the figure illustrates the two complementary strands of DNA, showing the AT and GC base pairs. Note that the orientation of the two strands is antiparallel. *See Sources & Acknowledgments.*

bases, **thymine** (T) and **cytosine** (C). Nucleotides, each composed of a base, a phosphate, and a sugar moiety, polymerize into long polynucleotide chains held together by 5′-3′ phosphodiester bonds formed between adjacent deoxyribose units (Fig. 2-3A). In the human genome, these polynucleotide chains exist in the form of a double helix (Fig. 2-3B) that can be hundreds of millions of nucleotides long in the case of the largest human chromosomes.

The anatomical structure of DNA carries the chemical information that allows the exact transmission of genetic information from one cell to its daughter cells and from one generation to the next. At the same time, the primary structure of DNA specifies the amino acid sequences of the polypeptide chains of proteins, as described in the next chapter. DNA has elegant features that give it these properties. The native state of DNA, as elucidated by James Watson and Francis Crick in 1953, is a double helix (see Fig. 2-3B). The helical structure resembles a right-handed spiral staircase in which its two polynucleotide chains run in opposite directions, held together by hydrogen bonds between pairs of bases: T of one chain paired with A of the other, and G with C. The specific nature of the genetic information encoded in the human genome lies in the sequence of C's, A's, G's, and T's on the two strands of the double helix along each of the chromosomes, both in the nucleus and in mitochondria (see Fig. 2-1). Because of the complementary nature of the two strands of DNA, knowledge of the sequence of nucleotide bases on one strand automatically allows one to determine the sequence of bases on the other strand. The double-stranded structure of DNA molecules allows them to replicate precisely by separation of the two strands, followed by synthesis of two new complementary strands, in accordance with the sequence of the original template strands (Fig. 2-4). Similarly, when necessary, the base complementarity allows efficient and correct repair of damaged DNA molecules.

Structure of Human Chromosomes

The composition of genes in the human genome, as well as the determinants of their expression, is specified in the DNA of the 46 human chromosomes in the nucleus plus the mitochondrial chromosome. *Each human*